Technology Stack

Date	04 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	***Streamlit*** *(or* ***React.js*** *if building a full-scale app)*	*Streamlit allows rapid prototyping of data science apps entirely in Python. It provides a clean, responsive UI for users to input financial details and instantly view prediction results without heavy frontend overhead.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	***FastAPI*** *(Python)*	*FastAPI is high-performance, asynchronous, and natively handles Python data structures. It is ideal for serving Machine Learning models (via libraries like* `Scikit-learn` *or* `XGBoost`*) with minimal latency and automatic API documentation.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*PostgreSQL*	*Highly reliable relational database. Credit card application data is inherently structured (income, credit score, employment history), making ACID compliance and robust relational mapping crucial for security and consistency.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	***Docker + AWS (EC2 / ECS)***	*Docker containerizes the application to prevent "it works on my machine" issues. AWS provides scalable, secure cloud hosting to deploy the containerized backend API and frontend, ensuring high availability.*

5	**Version Control & CI/CD** (e.g., GitHub, GitLab, Jenkins)	***GitHub*** *&* ***GitHub Actions***	*GitHub enables seamless team collaboration and code tracking. GitHub Actions automates the CI/CD pipeline, automatically running linting, testing, and deploying the application to the cloud whenever updates are pushed*

6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	***Scikit-learn*** */* ***Joblib*** *&* ***Experian/Equifax Sandbox API*** *(Simulated)*	`Scikit-learn` *handles data preprocessing and prediction logic, while* `Joblib` *loads the pre-trained model. A credit scoring API simulation represents the integration of real-world credit history data required for a genuine approval decision*